HOUSE PRICE ESTIMATION

A supervised machine-learning workflow for estimating property prices

LINEAR REGRESSION

500 SYNTHETIC RECORDS

PROJECT BRIEF

PROGRAM

Machine Learning ? IIOT-4

COURSE

BCA (Artificial Intelligence), Section A

MENTOR

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OUTPUT

Estimated price in INR

Educational demonstration ? not a formal property valuation

PROJECT TEAM

The team behind the project

Student names and IDs retained from the original presentation.

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A data-driven estimate from property features

House prices vary with characteristics and market context; a model can learn patterns from labelled examples.

01 **Problem**

Manual estimates can differ. Similar-sized homes may have different prices due to location, age, layout and market conditions.

02 **Approach**

Supervised regression learns from records containing property inputs and known listed prices, then predicts unseen cases.

PROJECT OBJECTIVES

Prepare

Inspect and clean a structured dataset.

Explore

Review feature and price patterns.

Model

Encode categories and train Linear Regression.

Evaluate

Measure held-out prediction errors and discuss limits.

EXPECTED OUTCOME — **A reproducible demonstration that returns an estimated price with evaluation metrics.**

02 ? DATASET

A reproducible dataset for a controlled experiment

The report describes 500 synthetic records. Their purpose is to demonstrate the workflow, not represent a live market.

500	records
400	training rows
100	test rows
42	random state

INPUT FEATURES

Area_sqft ? Bedrooms ? Bathrooms ? Age_years ? Location_Category

TARGET

Price_INR

DATA UNDERSTANDING & EDA

Load CSV ? inspect rows, types and summary statistics ? check missing values, duplicates and valid numeric fields ? use descriptive statistics and area-versus-price plots to identify trends or unusual records.

Prepare features consistently before model fitting

The held-out test set evaluates records that were not used to fit the model.

01 Inspect & clean
Review the CSV, data types, missing values and duplicates.

?

02 Separate inputs
Use four numeric features and one categorical location field.

?

03 Encode category
OneHotEncoder turns location groups into indicator columns.

?

04 Split records
80% train and 20% test; random_state=42 for reproducibility.

?

05 Fit pipeline
Transform inputs and fit Linear Regression on training data.

SCIKIT-LEARN PIPELINE

ColumnTransformer + OneHotEncoder(handle_unknown="ignore") + LinearRegression

Numeric features are passed through; the pipeline reuses the same transformation during training and prediction.

From raw records to price predictions

Linear Regression is a transparent baseline for a continuous target such as listed house price.

DATA
CSV records

?

INSPECT
Pandas / EDA

?

TRANSFORM
Numeric + encoded location

?

TRAIN
Linear Regression

?

PREDICT
Estimated INR

LINEAR MODEL (CONCEPT)

Predicted price = intercept + ? (feature coefficient ? feature value)

WHY THIS BASELINE?

Simple to explain; coefficients show fitted relationships, not causal effects.

After fitting: predict test rows ? compare with known prices ? calculate metrics ? inspect diagnostic plots.

05 ? RESULTS

Reported test-set metrics

Values below are reported for the synthetic dataset and must not be interpreted as real-market accuracy.

MAE

?382,443

Average absolute prediction error; same units as price.

RMSE

?490,665

Square-root mean squared error; larger errors weigh more.

R?

0.985

Proportion of test-target variation explained by the fitted model.

INTERPRETATION

MAE/RMSE express error in INR. R? summarizes fit on this held-out sample; none alone proves generalization.

Test subset: 100 records ? Split: 80:20 ? random_state=42

Use plots to inspect patterns and errors

The report describes these diagnostic plots; no plot values or individual observations are inferred here.

01 **Area vs Listed Price**

Scatter plot: inspect the general size?price trend and spread; area alone does not explain every difference.

02 **Actual vs Predicted**

Compare test values with predictions; a 45? reference line indicates perfect agreement.

03 **Error Distribution**

Histogram of actual minus predicted prices; values near zero indicate smaller errors.

04 **Linear Coefficients**

Compare fitted feature weights; signs show model direction, but not causation.

READ WITH CARE **Visual patterns describe this synthetic sample; coefficients depend on units and encoding.**

An educational baseline?not a valuation service

A practical tool would need more representative data, richer features and careful validation.

LIMITATIONS

- Synthetic data, not verified market transactions
- Broad location categories; no exact neighbourhood
- Missing condition, amenities, floor, legal status and demand
- Linear Regression may miss nonlinear patterns
- Results can shift with data and market changes

FUTURE SCOPE

- Use larger, verified listing or transaction data
- Add locality, amenities and property condition
- Compare tree-based models with Linear Regression
- Use cross-validation and systematic evaluation
- Report prediction ranges; consider Streamlit / Flask

RESPONSIBLE USE **Treat any output as an estimate; validate locally and communicate uncertainty.**

CONCLUSION

A clear starting point for regression-based prediction

The project demonstrates the supervised-learning workflow: prepare and explore data, encode location, split records, train Linear Regression, predict held-out prices, and evaluate results.

TAKEAWAY

The synthetic results teach the method?not real-world market accuracy.

Real use requires representative data, validation and uncertainty.

REFERENCES

01 Scikit-learn User Guide ? Linear Models: scikit-learn.org/stable/modules/linear_model.html

02 Scikit-learn ? OneHotEncoder: scikit-learn.org/stable/modules/generated/sklearn.preprocessing.OneHotEncoder.html

03 Pandas documentation: pandas.pydata.org/docs/

04 Matplotlib documentation: matplotlib.org/stable/

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